Preface

This book teaches derivatives the way the subject is actually organized: instruments and the infrastructure that clears them, then options and their valuation, then volatility, credit, rates, and the places where models are asked to do more than they can. That sequence is not mine. It was established by John Hull in *Options, Futures, and Other Derivatives*, refined across eleven editions, and it is the order in which the subject is taught almost everywhere for the good reason that it works. A student who understands forward pricing before futures, futures before swaps, and binomial trees before Black–Scholes is a student who has never had to accept a result without proof. I have followed that sequence deliberately and I want to say so at the outset, because a reader who knows Hull's book will recognize the architecture and should not have to wonder whether the resemblance is accidental.

What follows is not a revision of that book, and it is not authorized by or affiliated with it. It is a new one, written because the subject moved. Professor Hull passed away on January 31, 2026, and the eleventh edition, dated 2022, is the last. The years around it were not quiet ones. LIBOR was replaced by overnight rates compounded in arrears, and with it the single discount curve that a generation of textbooks assumed. Collateral became the thing that determines what a derivative is worth, so that valuation adjustments moved from a specialist annex into the price itself. Volatility modeling absorbed a genuinely new idea in rough volatility, which changes how the path of volatility itself is modeled. Clearing houses became systemically central, and their margin models became a market risk in their own right. Commodities and crypto each acquired enough substance to deserve a chapter rather than a remark, and machine learning enough to deserve the online technical note that accompanies the book. *A book that stopped in 2022 would now mislead a practitioner about the mechanics of the instruments they trade every day*, and that is the gap this one is written to fill.

I owe Professor Hull a great deal more than a sequence, and I came to the subject through him. My first doctorate was in applied mathematics (stochastic control theory), and I spent the year after it as a post-doctoral fellow in the operations management group at the Faculty of Management, University of Toronto, now the Joseph L. Rotman School of Management. By the end of that year I had decided against a career in mathematics, without much idea of what else the training was for. Someone there told me that a professor named John Hull was well known in financial derivatives, and that a mathematical background was useful there. I bought his book, then in its second edition, and the subject turned out to be a great deal more interesting than a way of salvaging a training. He took me on as a doctoral student in finance at the University of Toronto. That copy was the second edition; this book is written after the eleventh.

Much of what I understand about how to explain a model, and about the difference between a model that fits and a model that is right, I learned from him. Readers who want the original statements of results this book uses will find his research papers in Appendix D, cited where the results are used, alongside the rest of the literature.

There is something further that no citation captures. More than any other book, his brought people into this subject. A great many of the practitioners and academics now working in derivatives first understood what a derivative was from its pages, and would say so without being asked; I am one of them. That is a rarer achievement than any particular result, and it is

the standard this book is written against rather than one it expects to meet.

Where this book departs from its predecessor, it does so on purpose, and the departures are the reason it exists.

It argues. A textbook can present a model and leave the reader to infer what it is worth. This one states what each model captures, where it fails, and how a practitioner would tell, often with the failure quantified. Chapter 21 measures how badly Heston misses the short-dated skew; Chapter 26 shows that a senior tranche's expected loss moves by a factor of twenty-eight for a tripling of the default rate. The claims are meant to be checkable, and where I make one I have tried to give the number that supports it.

Every plot is computed, not drawn. The plots are generated from the same formulas the text derives, so a curve cannot drift out of step with the equation beside it. Where a worked example runs long, the script that produced it is kept, so the figure can always be recomputed.

It is bilingual, and the Chinese edition is a superset rather than a translation. It covers everything here, on the same terms, and adds the domestic market where it differs (China's benchmark architecture, its bond futures conventions, its structured products), because a Chinese reader needs both the international convention and the local one, and needs to understand the reason for the gap.

And it takes market structure seriously as subject matter. Clearing, margin, collateral, auctions, and the definitions inside contracts are not preliminaries to the mathematics. They are where a large share of actual losses come from, and several chapters are built around episodes (2008, the 2020 dash for cash, Archegos, the LME nickel squeeze, Credit Suisse's additional tier 1) in which the mechanism mattered more than the model.

The mathematical level is deliberately unchanged from its predecessor's: accessible, and worked through with intuition and numbers rather than asserted.

Five practical notes follow on how the book is put together.

Elective material. A dagger on a chapter or section title (Chapters 14, 21, 28, 30 and 32, and sections of Chapters 20, 38 and 39) marks material aimed at readers with stronger quantitative preparation. The mark appears in the table of contents and in the running head as well as on the heading, so a path through the book can be planned from the contents alone. Elective material can be skipped without breaking continuity: an MBA reader can pass from Chapter 13 to Chapter 15 without Chapter 14, for instance. A financial-engineering reader should not skip it, and nothing in it is optional in the sense of being less settled or less used—it is harder, not softer. Section **??** maps the dependencies between chapters and is the place to start if you are assembling a path rather than reading straight through.

Technical notes. Derivations that would interrupt the main argument are collected in a companion set of technical notes, published separately and referenced in the text as [TN15], numbered to match the chapter that defers to them; TN41, on machine learning, stands alone and follows the last chapter. The main text states results and explains why they hold; the notes prove them. The two the argument leans on hardest are reproduced in Appendix A, so a reader working from the book alone is never stopped by one.

Examples and case studies. Every formula whose magnitude matters is carried through actual numbers, because a reader who has not seen a convexity adjustment expressed in basis points does not yet know whether it is worth computing. Several worked examples deliberately verify a result two ways, and those cross-checks are part of the method being taught rather than decoration. The case studies carry the episodes: Metallgesellschaft, Amaranth, Archegos, the LME nickel squeeze, and the rest.

Companion notebooks. Twenty-nine chapters (3, 9 and 13–39) come with a Python notebook that reproduces their numbers; the online note on machine learning, TN41, has one as well. The code is written out rather than delegated: a calculation usually left to a library is

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set down step by step, because seeing it is the point of reading it. No chapter depends on its notebook, since each is complete as it stands, but a reader who wants to change a parameter and watch what moves will get further with one open. The two editions run identical code; only the narrative and the printed labels are translated.

Boxed equations. A box around an equation marks a result that later chapters build on and that a reader should be able to reproduce from memory. A chapter has one or two at most, and many have none; intermediate steps, approximations and special cases are never boxed, however often they are used.

Errors that remain are mine. I would be grateful to hear of them.

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